

Overview — All Methodologies

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Overview — every methodology, in plain language

This document explains, without formulas and without unnecessary jargon, every statistical methodology used across the portfolio. The goal is simple: you don't need a statistics background to understand **what question each method answers** and **what it lets you conclude** — and, just as important, **what it doesn't**.

Each section follows the same shape: a concrete question, the idea behind the method, what the result means, and one important limit.

For a more detailed explanation of any of these methods, see [methodologies/](#) (intermediate level) or [complete/](#) (full technical level).

1. Welch's t-test

The question: two groups have different average values in a sample. Is that difference probably real, or could it just as easily have come from the random variation of who happened to end up in each sample?

The idea: if you repeatedly drew small samples from two populations that actually had the same mean, the sample means would still almost never come out identical. The test measures how unusual the observed difference is, given the sample sizes and the spread within each group — without requiring the two groups to have the same internal variability, which makes it safer than the classic (Student's) version when the groups differ in size or spread.

What the result means: a low p-value says a difference this large would be rare if the population means were actually equal. That's evidence in favor of a real difference — not proof that it's large or practically important.

What it doesn't tell you: why the difference exists, or whether it's caused by anything in particular.

2. Oaxaca-Blinder decomposition

The question: two groups earn different average wages. How much of that comes from the groups having, on average, different characteristics (years of education, say), and how much is left over even after accounting for those?

The idea: the method fits a statistical model relating wages to observable characteristics separately for each group, then asks: "if Group B had Group A's characteristics but kept being paid according to its own wage structure, what gap would we expect?" That splits the

gap into an **explained** part (difference in characteristics) and an **unexplained** part (what's left over).

What the result means: the unexplained part is whatever can't be attributed to the measured characteristics. It's often read as a possible signal of discrimination — but that conclusion needs much stronger assumptions than the method alone provides: there may be relevant characteristics that simply weren't measured.

What it doesn't tell you: the method doesn't prove causation or isolate discrimination — it decomposes an observed difference, it doesn't explain its causal origin.

3. RIF (Recentered Influence Function) decomposition

The question: Oaxaca-Blinder decomposes the difference in **means**. What if the gap between groups isn't the same everywhere in the distribution — what if it's larger among low earners, or among high earners?

The idea: instead of looking only at the mean, the method lets you decompose the gap at any point of the distribution — the median, say, or the 90th percentile. It does this by transforming each observation into a measure of “how much it contributes” to that specific point of the distribution, then applies the same Oaxaca-Blinder logic on top of that transformed measure.

What the result means: it reveals patterns the mean hides — a small gap in the middle of the distribution but a large one at the tails, for example.

What it doesn't tell you: each point of the distribution is estimated with its own margin of error, usually wider at the tails (where there's less data) — results there deserve more caution.

4. Inequality decomposition (Theil, Gini, weighted quantiles)

The question: how much of the total inequality in a variable (income, say) comes from differences **between** groups, and how much from differences **within** each group?

The idea: the Theil index has a useful mathematical property called exact decomposability: total inequality can be split, with no leftover, into a “between-groups” share and a “within-groups” share. The Gini index measures concentration within a group (how far that group is from a perfectly equal distribution). Weighted quantiles estimate, within each group, cutoffs like “what defines the richest 10% of that group.”

What the result means: it's common, in broad demographic splits like race or gender, for most of the total inequality to sit within each group — that doesn't diminish the importance of a between-group gap that's large and statistically significant; it just shows inequality has several sources at once.

What it doesn't tell you: a lower Gini in one group doesn't necessarily mean a “better situation” — it can simply reflect a distribution more compressed near the bottom.

5. Duncan dissimilarity index

The question: are two groups distributed similarly across different categories (occupations, say), or is one group concentrated in certain categories while the other sits in different ones?

The idea: the index essentially measures what fraction of one group would need to “switch categories” for the two distributions to become identical. A value of zero means identical distributions; a value of one (or 100%) means complete segregation.

What the result means: a high index indicates the groups rarely occupy the same categories — a signal of occupational segregation, for instance.

What it doesn’t tell you: the index doesn’t say why the segregation exists, nor does it distinguish voluntary concentration from access barriers or differences in qualifications.

6. Structural break test

The question: did a series over time change pattern at a specific, known moment — an event, a law, a crisis?

The idea: the method compares the series’ behavior before and after a pre-defined date, testing whether the two pieces look like they come from the same process or from different ones.

What the result means: a significant break is evidence that something changed at that moment — but if several events happened close together (a pandemic near a labor reform, say), the test alone can’t say which one caused the change.

What it doesn’t tell you: this is evidence of temporal correlation, not causation — and a test that looks at one pre-defined date is not a search for unknown breaks anywhere in the series.

7. Difference-in-differences

The question: did a policy or event have a causal effect on an affected group, compared to an unaffected one?

The idea: compares the change over time in the affected group with the change over time in the unaffected group. If the two groups were already following similar trajectories before the event (so-called “parallel trends”), the difference in changes can be attributed to the event. Placebo tests apply the same logic to fake dates — if an “effect” shows up even on a date with no real policy change, that’s a warning sign.

What the result means: when parallel trends hold and the design is well executed, the method supports a causal reading — stronger than a simple before/after comparison.

What it doesn’t tell you: “we didn’t find an effect” isn’t the same as “there is no effect” — a design with few data points, or a small sample, may simply lack the statistical power to detect a real effect.

8. Panel transition analysis

The question: do people in one group leave unemployment, or enter formal work, faster or slower than people in another group?

The idea: some household surveys interview the same person more than once over time. That makes it possible to track real individual transitions (unemployed → employed, say), instead of only comparing isolated snapshots of different moments.

What the result means: different transition rates between groups show real differences in mobility, not just differences in composition at a single moment.

What it doesn't tell you: matching the same person across interviews can fail (a household member changing, for instance) — the match success rate matters for how much you can trust the result.

9. A/B testing

The question: does a specific change (to a product, a process) actually improve the outcome that matters, or does the apparent improvement come from noise?

The idea: a group of people is randomly split in two — half see the new version, half stay on the old one — and the average outcome is compared between the two groups, with a significance test and a confidence interval around the estimated effect.

What the result means: because the split is random, a significant difference between the groups can be attributed to the change being tested, not to other differences between the people.

What it doesn't tell you: this only holds if the randomization actually worked — hence the importance of always checking that before trusting the result (see the next item).

10. Sample Ratio Mismatch (SRM) test

The question: did the split between the experiment's groups actually come out as planned?

The idea: if you planned a 50/50 split but a bug put 55% of people in one group and 45% in the other, that's a sign something broke in the randomization — and any effect measured after that point can't be trusted.

What the result means: a significant SRM is an alarm: stop and investigate before interpreting any result from the experiment.

What it doesn't tell you: the test detects that the ratio is wrong, it doesn't automatically say why — that requires separate investigation.

11. Balance checks and covariate adjustment

The question: were the two experiment groups actually similar before the change was applied?

The idea: characteristics that already existed before the experiment (age, usage history, say) are compared between the two groups. If they look similar, the randomization worked. Those same characteristics can also be used to “clean up” some of the noise in the effect estimate, making it more precise.

What the result means: balanced groups give more confidence that any difference in outcome came from the change being tested, not from a pre-existing difference between the groups.

What it doesn't tell you: balance on the measured characteristics doesn't guarantee balance on unmeasured ones.

12. Non-inferiority test

The question: did a safety or risk metric (fraud rate, say) not get worse beyond an acceptable limit?

The idea: unlike an ordinary test, which asks “is there any difference?”, this one asks only “did this metric not get worse than X?” — in a single direction.

What the result means: when the test passes, there’s evidence the metric stayed within the acceptable range. When it doesn’t pass, that doesn’t necessarily mean the metric got much worse — it means there isn’t enough evidence that it stayed within the agreed margin.

What it doesn’t tell you: choosing the acceptable margin is a business decision, not a statistical one — the test doesn’t say which margin is “right.”

13. Heterogeneity (interaction) testing and novelty-effect decay

The question: is the effect of a change the same for everyone, or larger for some subgroups? And does the effect hold over time, or is it just a “novelty effect” that fades?

The idea: a formal interaction test checks whether the difference in effect between subgroups is large enough not to be just sampling noise. Looking at the effect across the weeks after launch shows whether it decays.

What the result means: finding an effect concentrated in one specific subgroup changes the practical decision — you might roll out only to that subgroup, instead of to everyone.

What it doesn’t tell you: testing many subgroups raises the chance of finding a “significant” difference by chance alone — an isolated heterogeneity result deserves more caution than one confirmed by several pieces of evidence.

14. Cluster-robust standard errors

The question: what if observations within the same group (a store, a region, a survey sampling unit) aren’t actually independent of each other?

The idea: the ordinary standard-error calculation assumes each observation is independent. When several observations come from the same cluster and look similar to one another, treating them all as independent makes the standard error look smaller than it really is. The cluster adjustment corrects for that.

What the result means: with the adjustment, the confidence interval comes out wider (and more honest) than it would without it.

What it doesn’t tell you: with very few clusters, even the adjusted standard error can be unreliable — that’s where alternative methods (items 15 and 16) come in.

15. Randomization inference

The question: with very few groups in the experiment (few stores, few regions), how do you trust a p-value computed the traditional way?

The idea: instead of assuming a theoretical formula, the method reshuffles the treatment/control assignment across groups in all possible ways (or many of them), computing the effect under each reshuffling. The p-value comes from comparing the observed effect to this distribution of “what if the assignment had been different” effects.

What the result means: it’s a form of inference that doesn’t depend on assumptions about the data’s distribution — only on the randomness of the assignment itself.

What it doesn’t tell you: it requires that the treatment assignment actually was random.

16. Wild-cluster bootstrap

The question: another way to get reliable inference with few clusters.

The idea: instead of resampling whole clusters (which works poorly when there are few of them), the method resamples the model’s residuals, multiplying them by random signs (+1 or -1) per cluster, and rebuilds the test statistic many times to form a reference distribution.

What the result means: it tends to agree with randomization inference when both are applicable — using more than one method and seeing whether they agree is a way of checking how robust a conclusion is.

What it doesn’t tell you: it still relies on some assumptions about the fitted model, even though it’s more robust than the classic formula.

17. Statistical power and Minimum Detectable Effect (MDE)

The question: before running an experiment, what’s the smallest effect it actually has a real chance of detecting?

The idea: it depends on the sample size, the metric’s variability, and the desired confidence level. Calculating this beforehand avoids running an expensive experiment that was never going to be able to see an effect of the size that matters.

What the result means: a high MDE (relative to the effect you’d realistically expect) is a sign that the experiment, as designed, likely won’t give a useful answer.

What it doesn’t tell you: not detecting an effect in an experiment with a high MDE doesn’t mean there’s no effect — only that the design didn’t have the power to see one.

18. Spatial spillover (cannibalization)

The question: is the positive effect measured in a treated area real gain, or is it activity “stolen” from neighboring control areas?

The idea: control areas near treated areas are compared with control areas farther away — if the nearby ones perform worse, that suggests part of the apparent “effect” is just displacement, not net creation.

What the result means: it helps estimate the real net effect of an intervention, not just its apparent local effect.

What it doesn’t tell you: fully isolating spillover is hard — the method gives an approximate estimate, not an exact measurement.

19. Cohort / retention decomposition

The question: why did an aggregate metric suddenly change — what specifically is driving it?

The idea: instead of looking only at the aggregate number, the base is split into subgroups (entry cohorts, segments, channels) and compared to see how each one contributed to the change — often revealing that several overlapping causes, not just one, explain the total.

What the result means: it lets you act on the specific cause, instead of reacting to the aggregate symptom.

What it doesn't tell you: the decomposition is descriptive — showing that a subgroup “contributed more” doesn't by itself prove it was the root cause without further investigation.

20. Growth decomposition

The question: did a business's growth come from expanding into new places/products, or from real improvement in what already existed?

The idea: total growth is split into an expansion part (new markets, new stores) and a “comparable” part (growth in what already existed long enough for a fair comparison) — and then, within the comparable part, split further into how much came from price, how much from volume, and how much from a change in the mix of products sold.

What the result means: an impressive total-growth number can hide a mature base that's stagnant or even shrinking.

What it doesn't tell you: the decomposition doesn't say whether the expansion strategy was a good call — it only separates where the total number came from.

21. Poisson regression (counts, with fixed effects)

The question: what explains how many goals a team scores — its own attack strength, the opponent's defense strength, and whether it's playing at home or away?

The idea: for count variables (number of events, like goals), Poisson regression models that count as a function of explanatory factors. Fixed effects per team capture each team's specific strength, isolating the effect of interest (playing at home, say).

What the result means: the coefficient for the factor of interest, once properly transformed, becomes a multiplicative factor — “playing at home multiplies the expected number of goals by X,” already net of the strength of both teams involved.

What it doesn't tell you: the model assumes a specific relationship between the factors and the expected count — it's worth checking whether that assumption is reasonable for the data at hand.

22. Weighted least squares trend

The question: is a metric rising, falling, or stable over time, for each unit being compared (a team, a club, a region)?

The idea: a straight line is fit to the data over time, giving more weight to periods with more observations (more reliable ones). Instead of deciding “rising or falling” just by looking at whether the slope is positive or negative, the slope’s confidence interval is examined: if the whole interval is positive, the trend is rising; if entirely negative, it’s falling; if it crosses zero, it’s classified as stable (not enough evidence of a trend).

What the result means: this classification avoids declaring a “trend” on top of noise — it requires the evidence to be strong enough to rule out the stable hypothesis.

What it doesn’t tell you: with few time periods, the confidence interval tends to be wide, and most units end up classified as “stable” simply for lack of enough data to decide — that’s not the same as saying they truly don’t change.